

Progettazione di Applicazioni Mobili

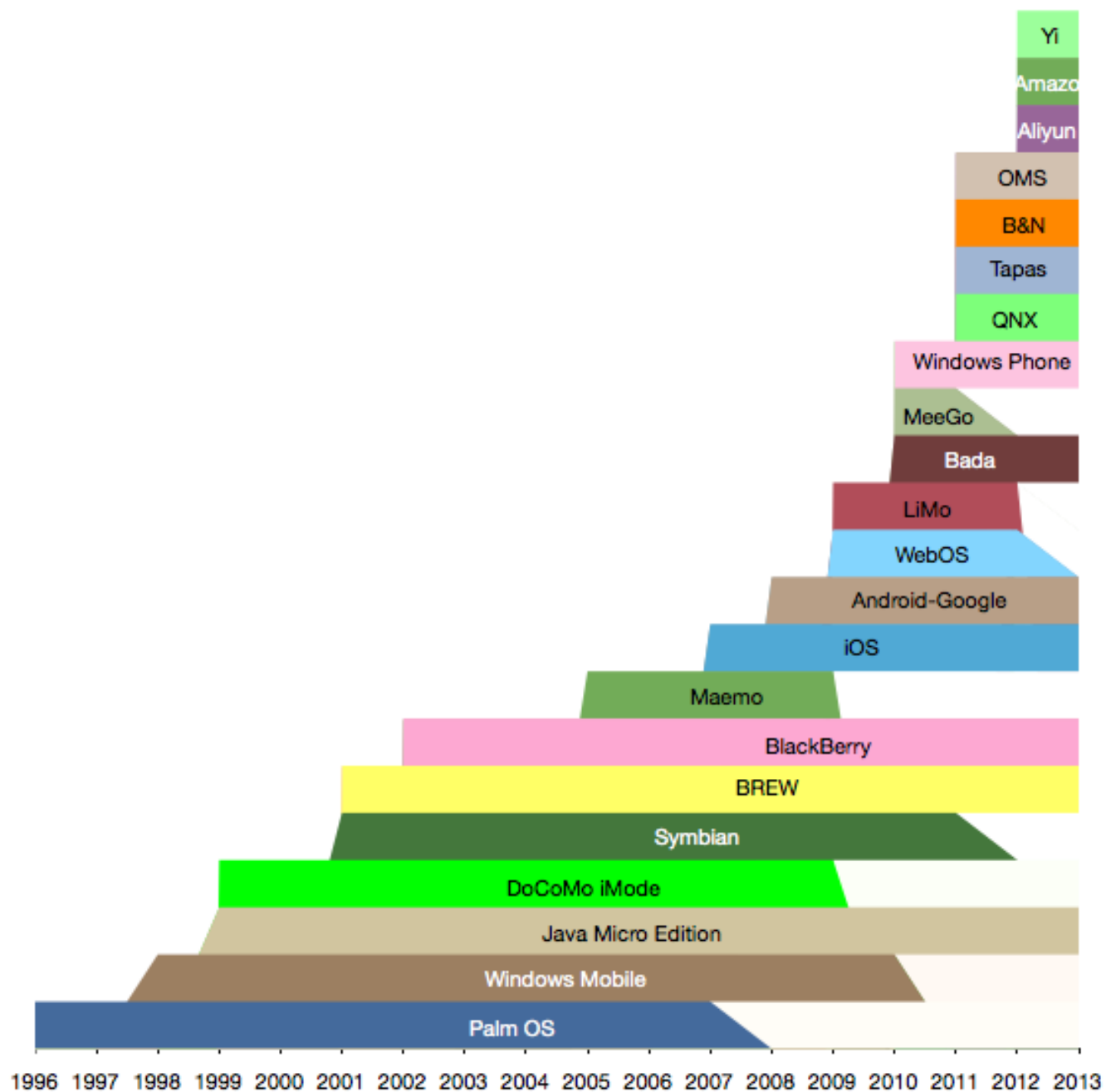
Smartphone platforms and development approaches

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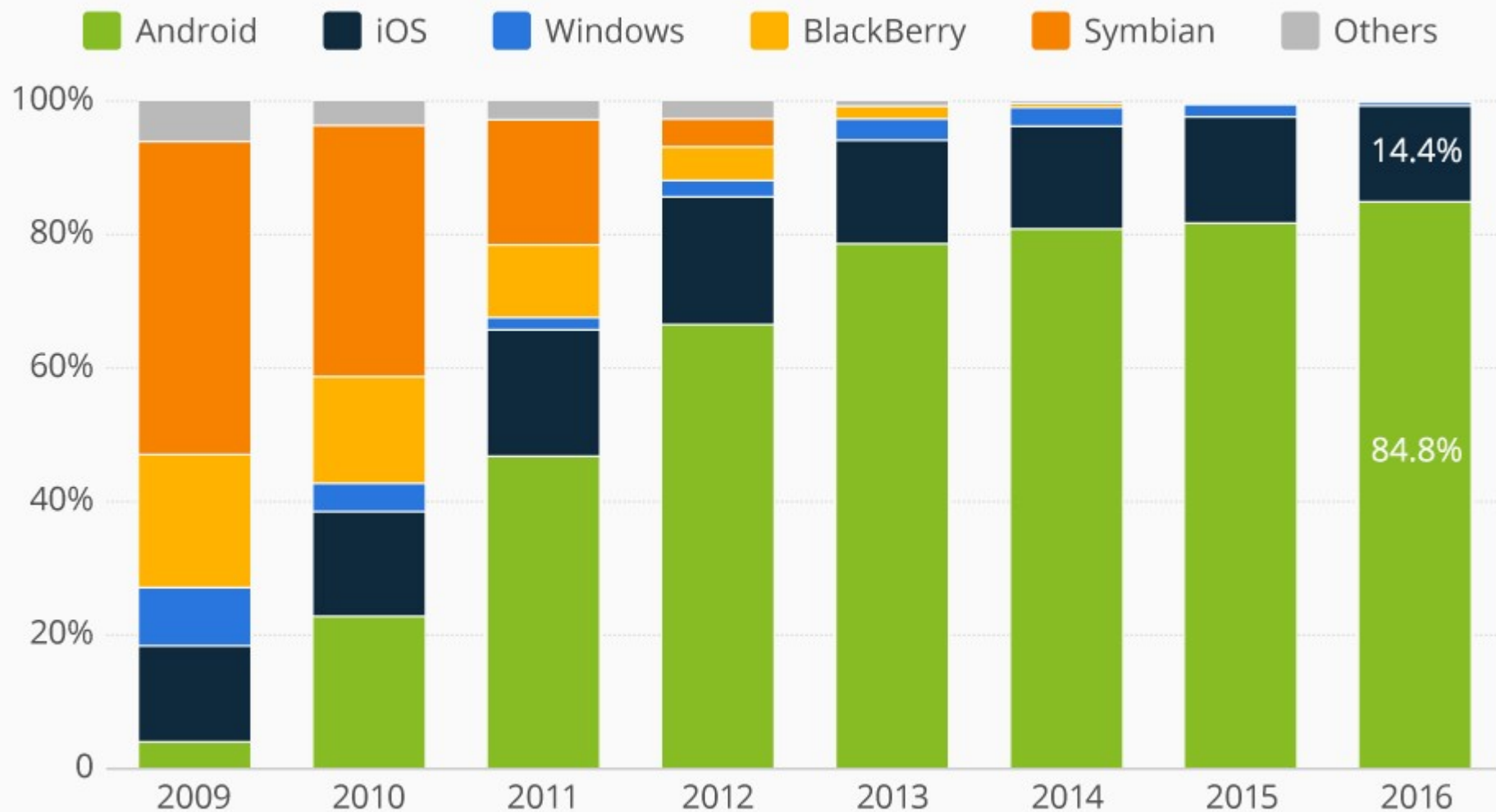




- 2012: Mozilla announces Firefox OS
- 2012: Sailfish OS launched by Jolla
- 2013 Canonical announces Ubuntu Touch, a version of the Linux distribution expressly designed for smartphones
- 2013: BlackBerry releases BlackBerry 10
- 2014: Windows Phone 8.1, iOS 8, Android 5.0
- 2015: iOS 9, Android 6.0, Windows 10 Mobile
- 2016: iOS 10, Android 7.0, Tizen 3.0
- 2017: iOS 11, Android 8.0
- 2018: iOS 12, Android 9.0, Tizen 4.0
- 2019: iOS 13 (+watchOS 6, iPad OS), Android 10
- 2020: iOS 14, Android 11
- ...

The Smartphone Platform War Is Over

Worldwide smartphone operating system market share (based on unit sales)







Android

73.9%

iOS

25.71%

Samsung

0.21%

Unknown

0.14%

KaiOS

0.02%

Linux

0.01%

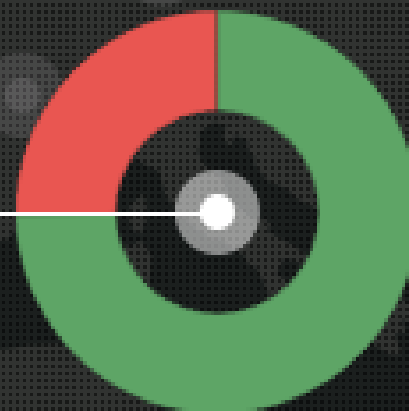
Mobile Operating System Market Share - August 2025

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Android	74.1%	
BlackBerry	0%	
iOS	25.3%	
Windows	0%	
Other	0.6%	

Compare



iOS/Android

- Unix-like operating systems (iOS derived from Darwin, Android derived from Linux)
- User interface based on the concept of direct manipulation and multi-touch gestures
- Set of UI design guidelines (Material Design for Android, Human Interface Guidelines for iOS)
- Native SDKs to support app development (access to almost all native features)
- Third-party native applications can be installed through app stores (App Store only for iOS)

iOS Development

- iOS SDK (free) available (since iOS 2.0, 2008) for app development in Objective-C (C variant, relatively low-level coding) or Swift (modern programming language with simpler syntax) using the Xcode development environment (free) on MacOSX
- In order to get applications to appear in the App Store, developers must enroll in the Apple Developer Program (\$99 per year)
 - Applications must be submitted to Apple for approval (Apple will take 15% of the revenue (up to \$1M), 30% over \$1M)

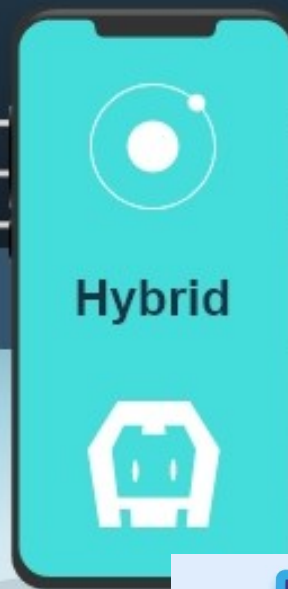
Android Development

- Android SDK available for app development in Java or Kotlin, using Google-developed libraries and Android Studio (free)
- Multiple supported development platforms: Linux, Mac OS X, or Windows
- Users can install apps directly (using APK files), or from alternative app markets (developers pay \$25 for registration to distribute on the Google Play Store, Google will take 15% of the revenue (up to \$1M), 30% over \$1M)

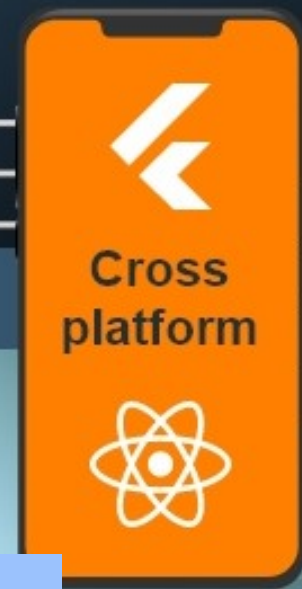
Native Vs. Hybrid Vs. Cross-platform



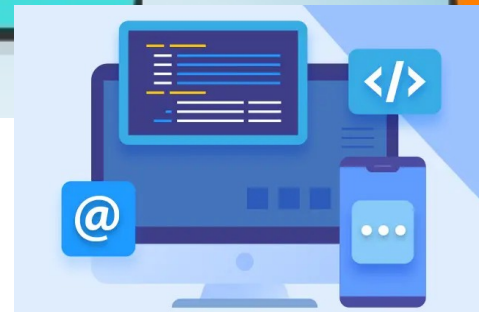
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Web app development

- A mobile web app is a mobile application that is hosted on the web and accessed from a browser on the mobile device
- Mobile web apps are developed using web technologies such as HTML5, CSS3 and JavaScript
- Web frameworks such as jQuery Mobile, Ext JS, Django can be used to simplify development of web-based mobile applications, providing a more consistent and comprehensive support across mobile browsers



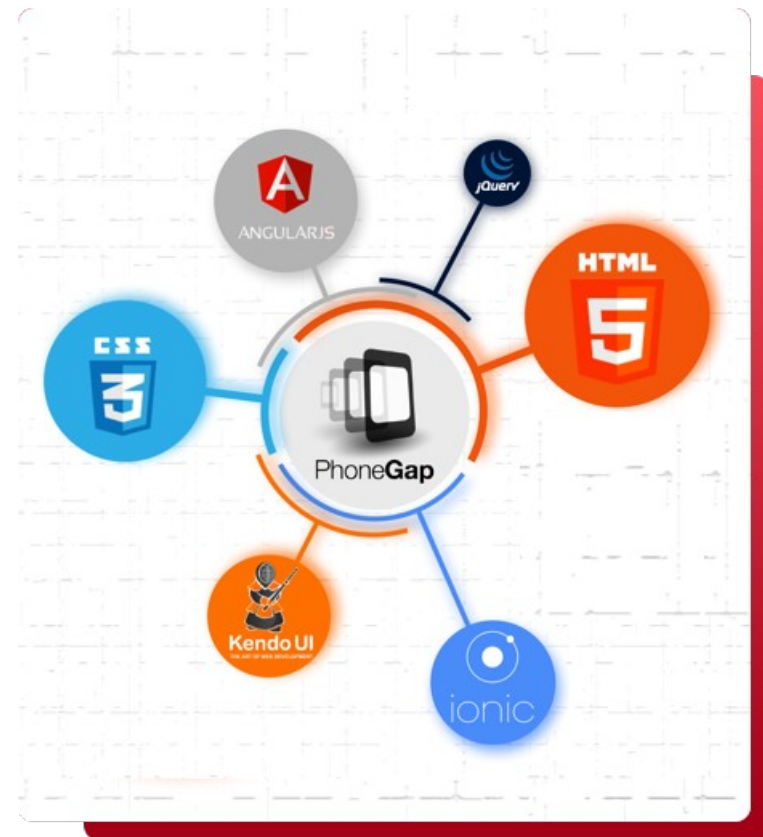
Native app development

- Native apps are built for specific platforms using the languages that the platform accepts and the specific development tools provided by the OS developer
 - Native development allows developers to directly access the full feature set of the selected operating system (including the UI system)



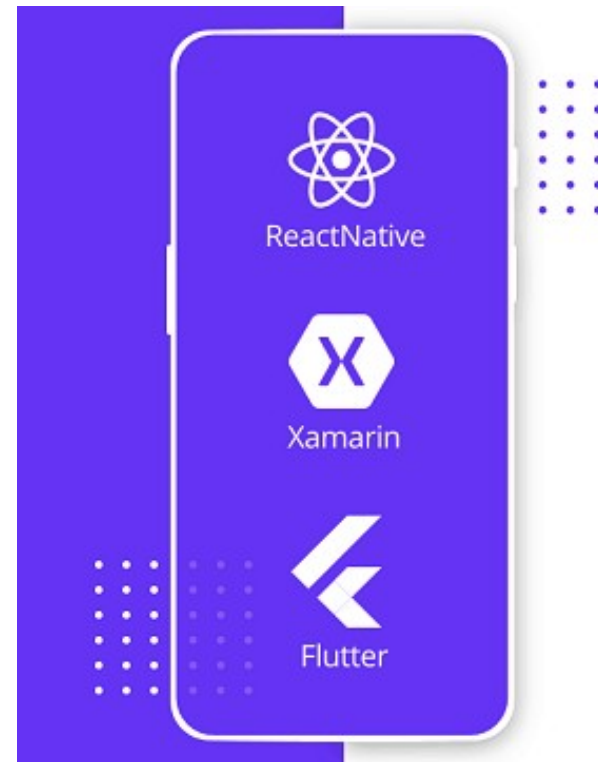
Hybrid app development

- Hybrid mobile applications run inside of a native container and use the device's web browser to display locally hosted HTML pages
- Hybrid apps contain a mixture of native app and mobile web app concepts. Typically the native code is provided by a framework and reveals a JavaScript API so that the app's JavaScript code can perform native functions.
- Well-known hybrid app frameworks: Ionic, Onsen UI, Mobile Angular UI, PhoneGap/Cordova, Kendo UI, Framework7



Cross-platform app development

- Cross-platform apps are native apps that are built using a single codebase, typically in a non-native language
- Popular cross-platform frameworks:
 - Appcelerator Titanium (js)
 - React Native (jsx)
 - Xamarin (C#)
 - Flutter (Dart)



	Native	Web	Hybrid - WebView	Cross-platform
Programming languages and Tech Stack	Different per platform (Java & Android SDK, Objective C & iOS SDK, C#, .NET & Windows 10 SDK)	HTML5, CSS3, JS	HTML5, CSS3, JS + extra SDKs on top (PhoneGap, Cordova, Ionic)	C# & .NET (Xamarin), JS (React Native, NativeScript, Titanium)
UI	Full native UI customization	Unified UI; limited options to mimic native look; most animations are absent	Unified UI; limited options to mimic native look	Fully native or unified
Developers availability	High to medium	High	Medium	Medium to low
Performance and UX	High, smooth animations and excellent UX.	Good to medium to poor, as it depends on browser compatibility, network and complexity of the code	Medium to poor - depends on complexity of the app	High - close to native
App Store distribution	Yes	No, hosted on company servers	Yes	Yes
Offline access	Yes	No	Yes, partial	Yes
Device API access	Full	Almost absent	Partial, however may result in bugs and crashes	Full or close to full
Code Reusability	No	Yes, 100%, however browser compatibility should be addressed	Yes, 100%	Up to 70-95% of the code
Development costs	High, as separate teams should be hired per platform	Low	Medium, as extra talents are required to use SDKs	Medium - extra skills and costs for UI tuning
Maintenance costs	High - different teams per platform	Medium	Low; medium to high - in long-term planning	Medium
Development time	Medium, however time-to-market may vary per platform	Quick	Quick	Medium
Maintenance complexity	Low to medium - depends on availability of developers	Low	Medium to high (grows with complexity of app and number of added features)	Low